

Differential Equations and Slope Fields

Lecture Notes

Topics: Modeling with DEs · Sign Analysis · Verifying Solutions · Slope Fields · Newton's Law of Cooling

Motivating Problem: A Falling Golf Ball

You drop a golf ball from a tall building. **What happens to its velocity over time?**

From everyday experience: it speeds up, but not without bound. The ball reaches a maximum speed (its **terminal velocity**) and stays there.

Why doesn't it keep accelerating? Two forces act on the ball during its fall:

- **Gravity:** $F_{\text{gravity}} = m g$, where $g \approx 9.8 \text{ m/s}^2$ is constant. We take downward as positive, so this force acts in the positive direction.
- **Air drag:** $F_{\text{drag}} = -b v$, where v is the current velocity and $b > 0$ is a drag constant. Drag opposes motion, hence the minus sign, and it grows with speed.

Note

Modeling drag as $F_{\text{drag}} = -b v$ (linear in v) is an oversimplification: for objects falling through air at everyday speeds, a better model for drag is $-b v^2$. The linear form is good enough to build intuition and yields a clean differential equation; getting the intuition for the v^2 case requires more advanced physics.

From Newton's Second Law to a Differential Equation

Newton's second law says $F_{\text{net}} = m a$. Summing the two forces:

$$m a = F_{\text{gravity}} + F_{\text{drag}} = m g - b v.$$

Acceleration is the time derivative of velocity, $a = \frac{dv}{dt}$, so

$$m \frac{dv}{dt} = m g - b v \implies \frac{dv}{dt} = g - \frac{b}{m} v.$$

Let $k = b/m$. The equation relating the ball's velocity to its acceleration:

$$\boxed{\frac{dv}{dt} = g - k v}$$

This is an equation that relates a quantity (v) to its own rate of change (dv/dt). It does *not* give us $v(t)$ directly. The unknown here is the entire **function** $v(t)$, not a single number.

Differential Equations

Differential Equation

An equation relating an unknown function to one or more of its derivatives. The unknown is a **function**, not a number.

An **ordinary differential equation (ODE)** involves derivatives with respect to a single independent variable. The *order* of the equation is the highest derivative that appears.

The general form of a first-order ODE is

$$\frac{dy}{dt} = f(t, y),$$

where f is some function of the independent variable t and the unknown y .

Autonomous ODE

A first-order ODE is **autonomous** if the right-hand side does not depend explicitly on t :

$$\frac{dy}{dt} = f(y).$$

The slope at any point depends only on the current value of y , not on the current time.

The golf-ball equation $\frac{dv}{dt} = g - kv$ is autonomous: only the current velocity v shows up on the right, not the current time t .

Initial Conditions

A DE typically has infinitely many solutions, one for each starting condition. An **initial condition** pins down a specific solution by saying:

$$y(t_0) = y_0$$

which means “at time t_0 , the function takes the value y_0 .” For the golf ball, dropping it from rest means $v(0) = 0$.

Sign Analysis: Reading a DE Without Solving It

Before trying to find a formula for $v(t)$, we can already say a lot about the long-run behavior of the golf ball just by reading off the sign of $\frac{dv}{dt}$.

Where Does the System Settle?

The velocity is unchanging exactly when $\frac{dv}{dt} = 0$. For the golf-ball equation:

$$g - kv = 0 \implies v = \frac{g}{k}.$$

This is the **terminal velocity**. With $g = 9.8 \text{ m/s}^2$ and (say) $k = 0.31 \text{ s}^{-1}$:

$$v_{\text{terminal}} = \frac{9.8}{0.31} \approx 31.6 \text{ m/s}.$$

What About Other Starting Speeds?

Check the three cases:

Current v	Sign of dv/dt	Behavior
$v < g/k$	$g - kv > 0$	accelerating (speeding up toward g/k)
$v = g/k$	0	equilibrium (terminal velocity)
$v > g/k$	$g - kv < 0$	decelerating (slowing down toward g/k)

From every starting velocity, the system is driven toward the same value $v = g/k$. We call this a **fixed point** or **equilibrium**, and because nearby solutions move *toward* it, this fixed point is **stable**.

Note

Sign analysis gives us qualitative behavior (the long-run story, the location and stability of equilibria) without producing a formula. It cannot tell us the value of v at a specific time like $t = 2$ seconds.

Verifying a Solution to a DE

Before we learn how to *produce* a solution, let's nail down how to *check* one. Given a DE and a candidate function, we want to verify that the function satisfies the equation.

Verifying a candidate solution

Given a DE and a candidate $y(t)$:

1. Differentiate $y(t)$ to get $y'(t)$.
2. Substitute $y(t)$ and $y'(t)$ into the DE.
3. Check whether the two sides agree for all t .

Does $y(t) = e^{3t}$ solve $y' = 3y$?

Step 1: differentiate: $y'(t) = 3e^{3t}$.

Step 2: substitute into $y' = 3y$:

$$\underbrace{3e^{3t}}_{y'(t)} \stackrel{?}{=} 3 \underbrace{e^{3t}}_{y(t)} = 3e^{3t}. \quad \checkmark$$

Both sides equal $3e^{3t}$ for every t , so $y(t) = e^{3t}$ is a solution.

A Whole Family of Solutions

Try the same DE with a constant multiple: $y(t) = C e^{3t}$ for any constant C .

$y'(t) = 3C e^{3t}$, and $3y = 3 \cdot C e^{3t} = 3C e^{3t}$. Match for every t .

So $y(t) = C e^{3t}$ solves $y' = 3y$ for *any* value of C . The differential equation describes an entire family of curves; the initial condition picks out which one.

Note

Throughout this course we write y when we really mean $y(t)$, the unknown function. Diff EQs almost always involve more than just plugging in a single value of t . Keep in mind that y varies with t even when the notation gets compact.

Slope Fields

A first-order ODE $\frac{dy}{dt} = f(t, y)$ assigns a slope to every point (t, y) in the plane. Drawing those slopes as little segments produces a **slope field**, the geometric picture of the DE.

Slope Field and Solution Curve

The **slope field** of a first-order ODE is a grid of short line segments. At each point (t, y) the segment has slope $\frac{dy}{dt} = f(t, y)$.

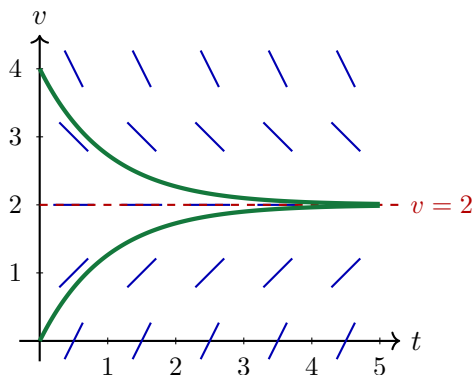
A **solution curve** (or **trajectory**) is any curve in the (t, y) -plane whose tangent at every point matches the slope field. An initial condition selects a single trajectory.

A Toy Example: $v' = 2 - v$

We pick an autonomous example that's easy to do arithmetic on. At each value of v , the slope $v' = 2 - v$ is:

v	$v' = 2 - v$
0	2
1	1
2	0
3	-1
4	-2

Because the equation is autonomous, the slope depends only on v (not on t), so each horizontal row of the slope field looks the same. The fixed point sits at $v = 2$. Trajectories above $v = 2$ slope downward; trajectories below slope upward. Everything funnels toward $v = 2$.



Newton's Law of Cooling

A cup of coffee at 90°C (194°F) is set down in a room at $T_a = 22^\circ\text{C}$ (72°F). **How does its temperature change over time?**

From experience: it cools, fast at first then slowing, and stops near room temperature. The rate of cooling depends on the gap between the coffee temperature T and the room temperature T_a .

Newton's Law of Cooling

The rate of temperature change of an object is proportional to the difference between its temperature and the ambient temperature:

$$\frac{dT}{dt} = -k(T - T_a)$$

where $k > 0$ is a rate constant (units: 1/time) and T_a is the ambient temperature.

Sign Analysis (Same Story as the Golf Ball)

Set $\frac{dT}{dt} = 0$: $T - T_a = 0$ gives $T = T_a$. The equilibrium is the room temperature itself. Check the three regimes:

Current T	Sign of dT/dt	Behavior
$T > T_a$ (hot coffee)	$-k(\text{positive}) < 0$	cools toward T_a
$T = T_a$	0	equilibrium
$T < T_a$ (cold drink)	$-k(\text{negative}) > 0$	warms toward T_a

Every starting temperature relaxes toward T_a . The slope field would look very similar to the golf-ball example: arrows funneling onto the equilibrium line. With $k = 0.05 \text{ min}^{-1}$, the coffee takes roughly 90 minutes to come close to room temperature.

Verifying a Candidate Formula

Claim: $T(t) = 22 + 68 e^{-kt}$ solves $\frac{dT}{dt} = -k(T - 22)$ with initial condition $T(0) = 90$.

Verifying the cooling formula

Step 1: differentiate $T(t) = 22 + 68 e^{-kt}$:

$$T'(t) = -68 k e^{-kt}.$$

Step 2: substitute into the right-hand side $-k(T - 22)$:

$$-k(T - 22) = -k(22 + 68 e^{-kt} - 22) = -k \cdot 68 e^{-kt} = -68 k e^{-kt}.$$

Step 3: both sides equal $-68 k e^{-kt}$. ✓

Check the initial condition: $T(0) = 22 + 68 e^0 = 22 + 68 = 90$. ✓

Check the long-time limit: as $t \rightarrow \infty$, $e^{-kt} \rightarrow 0$, so $T(t) \rightarrow 22$. The coffee approaches room temperature. ✓

We verified that this formula works. We haven't yet learned how anyone would have *found* it from scratch. That's the subject of the next lesson.